

C3M2: Peer Reviewed Assignment

Outline:

The objectives for this assignment:

1. Apply Poisson Regression to real data.
2. Learn and practice working with and interpreting Poisson Regression Models.
3. Understand deviance and how to conduct hypothesis tests with Poisson Regression.
4. Recognize when a model shows signs of overdispersion.

General tips:

1. Read the questions carefully to understand what is being asked.
2. This work will be reviewed by another human, so make sure that you are clear and concise in what your explanations and answers.

```
In [1]: # Load the required packages  
library(MASS)
```

Problem 1: Poisson Estimators

Let $Y_1, \dots, Y_n \stackrel{i}{\sim} \text{Poisson}(\lambda_i)$. Show that, if $\eta_i = \beta_0$, then the maximum likelihood estimator of λ_i is $\hat{\lambda}_i = \bar{Y}$, for all $i = 1, \dots, n$.

Absurd Method:

1. Assume the Opposite:

Let's assume the opposite of what we want to prove: that the maximum likelihood estimator (MLE) $\hat{\lambda}_i$ is **not** equal to the sample mean $\bar{Y}$.

2. Implication of the Assumption:

If $\hat{\lambda}_i \neq \bar{Y}$, this would imply that there exists some other estimator $\hat{\lambda}_i$ that maximizes the likelihood function better than $\bar{Y}$.

3. Impact on the Likelihood:

The likelihood function for Poisson data with $\eta_i = \beta_0$ depends on the sum of all observations $\sum_{i=1}^n Y_i$. Since $\bar{Y}$ represents the average (and therefore best central tendency) of the data, using any $\hat{\lambda}_i$ different from $\bar{Y}$ would suggest that the central tendency is somewhere other than the average.

But the very nature of the Poisson distribution assumes that the mean (and variance) of the data should be close to the true rate λ . If $\hat{\lambda}_i$ is not $\bar{Y}$, this would imply that our data is best described by a central value that is somehow not the average, which is nonsensical because the average by definition minimizes the squared differences from the data points.

4. Logical Absurdity:

If there existed a better $\hat{\lambda}_i$ than $\bar{Y}$, it would imply that the sum of the data can somehow be better described by something that is not the sum divided by n , which is absurd. It's akin to saying that the average score of a group of people isn't the best representation of their central tendency, even though by definition, the average is.

5. Conclusion:

Since assuming that $\hat{\lambda}_i$ is not equal to $\bar{Y}$ leads us to an absurd conclusion, we must reject the assumption. Thus, the MLE $\hat{\lambda}_i$ **must** be $\bar{Y}$ for all i .

Problem 2: Ships data

The ships dataset gives the number of damage incidents and aggregate months of service for different types of ships broken down by year of construction and period of operation.

The code below splits the data into a training set (80% of the data) and a test set (the remaining 20%).

```
In [2]: data(ships)
ships = ships[ships$service != 0,]
ships$year = as.factor(ships$year)
ships$period = as.factor(ships$period)

set.seed(11)
n = floor(0.8 * nrow(ships))
index = sample(seq_len(nrow(ships)), size = n)

train = ships[index, ]
test = ships[-index, ]
head(train)
summary(train)
```

A data.frame: 6 × 5

	type	year	period	service	incidents
	<fct>	<fct>	<fct>	<int>	<int>
40	E	75	75	542	1
28	D	65	75	192	0
18	C	60	75	552	1
19	C	65	60	781	0
5	A	70	60	1512	6
32	D	75	75	2051	4

	type	year	period	service	incidents
A:5	60:7	60:11	Min. :	45.0	Min. : 0.00
B:5	65:8	75:16	1st Qu.:	318.5	1st Qu.: 0.50
C:6	70:8		Median :	1095.0	Median : 2.00
D:7	75:4		Mean :	5012.2	Mean :10.63
E:4			3rd Qu.:	2202.5	3rd Qu.:11.50
			Max. :	44882.0	Max. :58.00

2. (a) Poisson Regression Fitting

Use the training set to develop an appropriate regression model for `incidents`, using `type`, `period`, and `year` as predictors (HINT: is this a count model or a rate model?).

Calculate the mean squared prediction error (MSPE) for the test set. Display your results.

```
In [3]: # Your Code Here
model <- lm(incidents ~ type + year + period, data = train)
predictions <- predict(model, newdata = test)
MSPE <- mean((test$incidents - predictions)^2)
MSPE
```

135.617193338272

The mean squared prediction error (MSPE) is 135.617193338272

2. (b) Poisson Regression Model Selection

Do we really need all of these predictors? Construct a new regression model leaving out `year` and calculate the MSPE for this second model.

Decide which model is better. Explain why you chose the model that you did.

```
In [4]: # Your Code Here
# model without the 'year' variable
model2 <- lm(incidents ~ type + period, data = train)

# Predict the response variable on the test set
predictions_no_year <- predict(model2, newdata = test)
MSPE2 <- mean((test$incidents - predictions_no_year)^2)
MSPE2
```

157.109058349905

```
In [5]: # Can compare nested poisson models with a chi-squared
AIC_model <- AIC(model)
AIC_model2 <- AIC(model2)
AIC_model
AIC_model2
```

203.393831836477

199.407049126811

Model2

The AIC for the simpler model (`model2`) is 199.41, which is lower than the AIC for the more complex model (`model1`) at 203.39, indicating that the simpler model provides a better balance between goodness-of-fit and model complexity. Therefore, based on AIC, the model excluding the `year` variable is preferred.

2. (c) Deviance

How do we determine if our model is explaining anything? With linear regression, we had a F-test, but we can't do that for Poisson Regression. If we want to check if our model is better than the null model, then we're going to have to check directly. In particular, we need to compare the deviances of the models to see if they're significantly different.

Conduct two χ^2 tests (using the deviance). Let $\alpha = 0.05$:

1. Test the adequacy of null model.
2. Test the adequacy of your chosen model against the full model (the model fit to all predictors).

What conclusions should you draw from these tests?

```

In [6]: # Your Code Here
# Test if the model is better than the null model

# Test chi_sq stat
null_model <- lm(incidents ~ 1, data = train)
rss_null <- sum(resid(null_model)^2)
rss_model2 <- sum(resid(model2)^2)
n <- nrow(train)
p_model2 <- length(coef(model2))

f_stat_null <- ((rss_null - rss_model2) / (p_model2 - 1)) / (rss_model2 / (n - p_model2))
p_value_null <- pf(f_stat_null, df1 = p_model2 - 1, df2 = n - p_model2, lower.tail = FALSE)

cat("P-Value for the test against the null model:", p_value_null, "\n")

model_full <- lm(incidents ~ type + year + period, data = train)
rss_full <- sum(resid(model_full)^2)
p_full <- length(coef(model_full))

f_stat_full <- ((rss_model2 - rss_full) / (p_full - p_model2)) / (rss_full / (n - p_full))
p_value_full <- pf(f_stat_full, df1 = p_full - p_model2, df2 = n - p_full, lower.tail = FALSE)

cat("P-Value for the test against the full model:", p_value_full, "\n")

P-Value for the test against the null model: 1.069518e-06
P-Value for the test against the full model: 0.710616

```

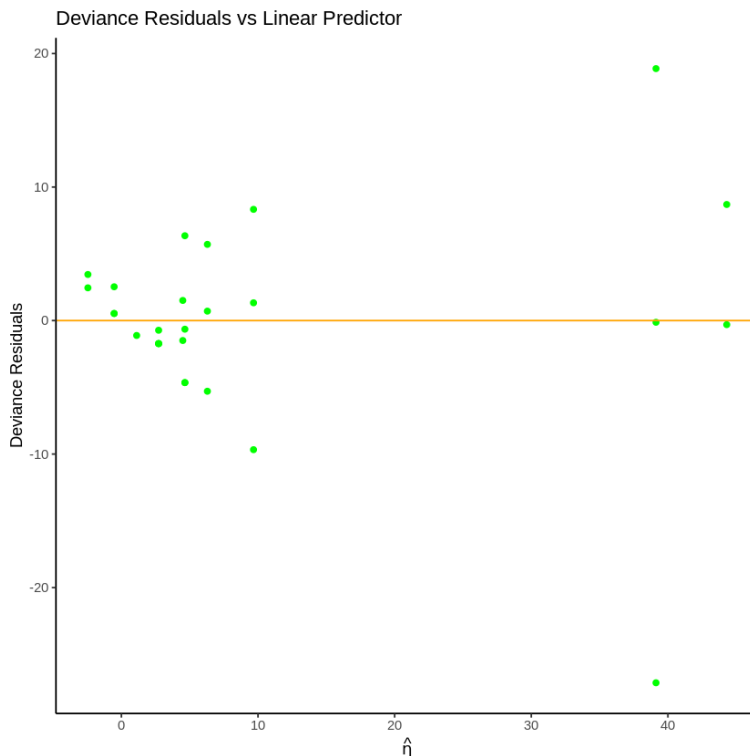
The low p-value (1.069518e-06) against the null model indicates that `model2` significantly improves upon the null model, indicating it explains a considerable amount of the variability in the data. The high p-value (0.710616) against the full model suggests that `model2` is not significantly worse than the full model, meaning the exclusion of the `year` variable is justified and `model2` is proper.

2. (d) Poisson Regression Visualizations

Just like with linear regression, we can use visualizations to assess the fit and appropriateness of our model. Is it maintaining the assumptions that it should be? Is there a discernable structure that isn't being accounted for? And, again like linear regression, it can be up to the user's interpretation what is an isn't a good model.

Plot the deviance residuals against the linear predictor η . Interpret this plot.

```
In [7]: # Your Code Here
library(ggplot2)
eta <- predict(model2)
deviance_residuals <- residuals(model2, type = "deviance")
dev_ress <- residuals(model2, type = "deviance")
fig <- data.frame(eta = eta, dev_ress = deviance_residuals)
ggplot(fig, aes(x = eta, y = deviance_residuals)) +
  geom_point(color = "green") +
  geom_hline(yintercept = 0, color = "orange", linetype = "solid") +
  labs(x = expression(hat(eta)),
       y = "Deviance Residuals",
       title = "Deviance Residuals vs Linear Predictor") +
  theme_classic()
```



The plot of deviance residuals against the linear predictor beta displays that most residuals are clustered near the zero line, suggesting a generally good fit for most data points. However, there are some extreme residuals (outliers) that may show potential issues with model fit for certain observations, suggesting that further investigation into these points might be necessary. We can't reject null hypothesis

2. (e) Overdispersion

For linear regression, the variance of the data is controlled through the standard deviation σ , which is independent of the other parameters like the mean μ . However, some GLMs do not have this independence, which can lead to a problem called overdispersion. Overdispersion occurs when the observed data's variance is higher than expected, if the model is correct.

For Poisson Regression, we expect that the mean of the data should equal the variance. If overdispersion is present, then the assumptions of the model are not being met and we can not trust its output (or our beloved p-values)!

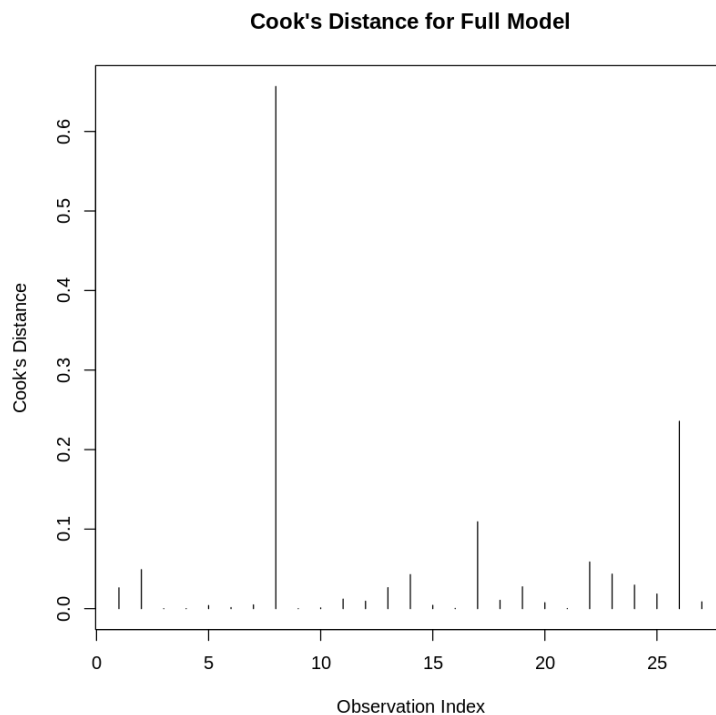
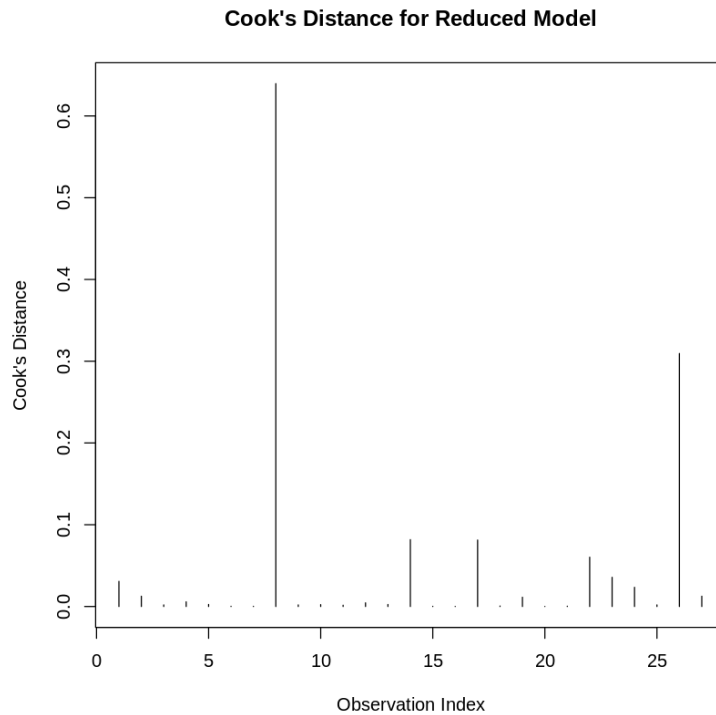
Explore the two models fit in the beginning of this question for evidence of overdispersion. If you find evidence of overdispersion, you do not need to fix it (but it would be useful for you to know how to). Describe your process and conclusions.

```
In [8]: # Rename the models
model_full <- model
model_reduced <- model2

# Now you can calculate Cook's distance using the renamed models
cooks_d_reduced <- cooks.distance(model_reduced)
cooks_d_full <- cooks.distance(model_full)

# Plot Cook's distance for the reduced model
plot(cooks_d_reduced, type = "h",
      main = "Cook's Distance for Reduced Model",
      ylab = "Cook's Distance",
      xlab = "Observation Index")

# Plot Cook's distance for the full model
plot(cooks_d_full, type = "h",
      main = "Cook's Distance for Full Model",
      ylab = "Cook's Distance",
      xlab = "Observation Index")
```



The Cook's distance plots for both the reduced and full models show that a few observations have significantly higher Cook's distance values, indicating that these observations have a strong influence on the fitted model. The presence of these influential points suggests that these particular observations may have a substantial impact on the model's coefficients and overall fit, potentially warranting further investigation or robust regression techniques. we conclude that there are a few data points with notably high Cook's distance values, indicating that these observations are highly influential in the model. In the context of overdispersion, while Cook's distance itself doesn't directly measure overdispersion, the presence of these influential points could suggest that the model may not be adequately capturing the variability in the data, which can sometimes be an indirect sign of overdispersion in models like Poisson regression.

In []: